# U23AI016
# BDA
# Assignment 9

**Exercise 1:** The Air Quality "Mini-ETL"
Goal: Clean a dataset and visualize correlation using a regression line.

- Dataset: Built-in airquality dataset. File Name: air_quality_lab.csv

Columns: Ozone (ppb), Solar_R (langley), Wind (mph), Temp (F), Month (1-12), Day (1-31).

- Tasks:

1. Handle missing values (NA) specifically in the Ozone and Solar.R columns.
2. Use the mutate function to create a "Temperature Category" (Cold, Moderate, Hot).
3. Visualization: Create a scatter plot of Solar.R vs. Ozone. Map the "Temperature Category" to the color aesthetic.
4. Requirement: Add a geom_smooth() layer to show the trend.

Ans :

R

```
user@DESKTOP-0OTJMH0:~/bda_lab/Lab_9$ R

R version 4.3.3 (2024-02-29) -- "Angel Food Cake"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
```

1) Handle missing values (NA) specifically in the Ozone and Solar.R columns.

Loading the dataset and viewing

```
> data("airquality")
> head(airquality)
  Ozone Solar.R Wind Temp Month Day
1    41     190  7.4   67     5   1
2    36     118  8.0   72     5   2
3    12     149 12.6   74     5   3
4    18     313 11.5   62     5   4
5    NA      NA 14.3   56     5   5
6    28      NA 14.9   66     5   6
```

Checking missing values

```
> colSums(is.na(airquality))
  Ozone Solar.R    Wind    Temp   Month     Day
     37       7       0       0       0       0
```

Handle missing values

```
> airquality$Ozone[is.na(airquality$Ozone)] <- mean(airquality$Ozone, na.rm = TRUE)
airquality$Solar.R[is.na(airquality$Solar.R)] <- mean(airquality$Solar.R, na.rm = TRUE)

clean_data <- airquality
> colSums(is.na(clean_data))
head(clean_data)
```

Output :

```
  Ozone Solar.R    Wind    Temp   Month      Day
      0       0       0       0       0        0
     Ozone  Solar.R Wind Temp Month Day
1 41.00000 190.0000  7.4   67     5   1
2 36.00000 118.0000  8.0   72     5   2
3 12.00000 149.0000 12.6   74     5   3
4 18.00000 313.0000 11.5   62     5   4
5 42.12931 185.9315 14.3   56     5   5
6 28.00000 185.9315 14.9   66     5   6
> |
```

2) Use the mutate function to create a "Temperature Category" (Cold, Moderate, Hot).

Installing dataset

```
> install.packages("dplyr")
Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)
Warning in install.packages("dplyr") :
  'lib = "/usr/local/lib/R/site-library"' is not writable
Would you like to use a personal library instead? (yes/No/cancel) yes
Would you like to create a personal library
‘/home/user/R/x86_64-pc-linux-gnu-library/4.3’
to install packages into? (yes/No/cancel) yes
also installing the dependencies ‘utf8’, ‘pkgconfig’, ‘withr’, ‘cli’, ‘generics’, ‘glue’, ‘lifecycle’, ‘magrittr’, ‘pillar’, ‘R6’, ‘r
lang’, ‘tibble’, ‘tidyselect’, ‘vctrs’

trying URL 'https://cloud.r-project.org/src/contrib/utf8_1.2.6.tar.gz'
Content type 'application/x-gzip' length 243856 bytes (238 KB)
==================================================
downloaded 238 KB

trying URL 'https://cloud.r-project.org/src/contrib/pkgconfig_2.0.3.tar.gz'
Content type 'application/x-gzip' length 6080 bytes
==================================================
downloaded 6080 bytes

trying URL 'https://cloud.r-project.org/src/contrib/withr_3.0.2.tar.gz'
Content type 'application/x-gzip' length 103240 bytes (100 KB)
==================================================
downloaded 100 KB

trying URL 'https://cloud.r-project.org/src/contrib/cli_3.6.5.tar.gz'
Content type 'application/x-gzip' length 640240 bytes (625 KB)
```

Loading dataset

```
> library(dplyr)

Attaching package: ‘dplyr’

The following objects are masked from ‘package:stats’:

    filter, lag

The following objects are masked from ‘package:base’:

    intersect, setdiff, setequal, union

>
```

Creating new column and verifying

```
> clean_data <- clean_data %>%
  mutate(
    Temp_Category = case_when(
      Temp < 70 ~ "Cold",
      Temp >= 70 & Temp <= 85 ~ "Moderate",
      Temp > 85 ~ "Hot"
    )
  )
> head(clean_data)
     Ozone  Solar.R Wind Temp Month Day Temp_Category
1 41.00000 190.0000  7.4   67     5   1          Cold
2 36.00000 118.0000  8.0   72     5   2      Moderate
3 12.00000 149.0000 12.6   74     5   3      Moderate
4 18.00000 313.0000 11.5   62     5   4          Cold
5 42.12931 185.9315 14.3   56     5   5          Cold
6 28.00000 185.9315 14.9   66     5   6          Cold
>
```

Checking category counts

```
> table(clean_data$Temp_Category)
```

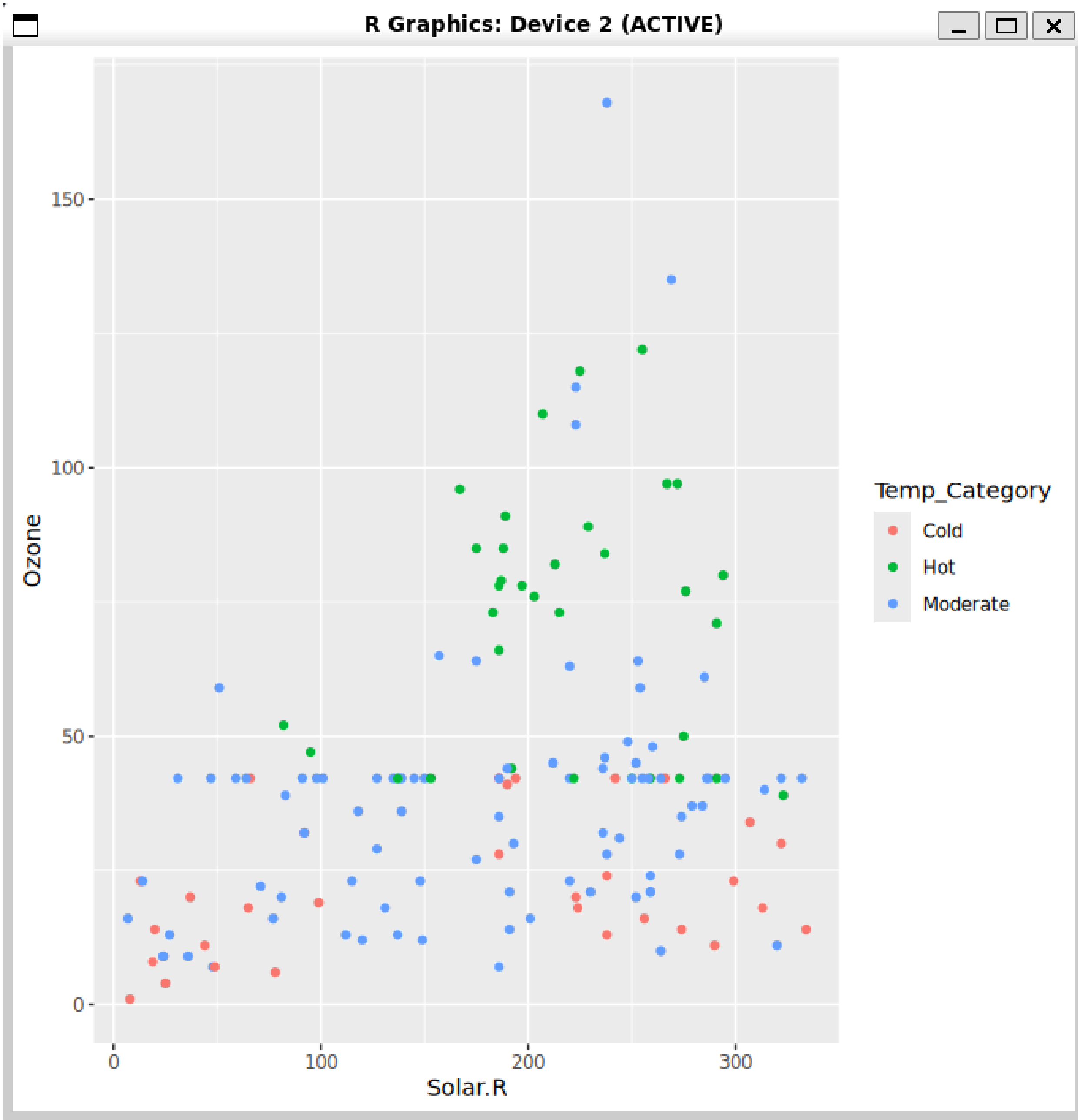
Output :

```
    Cold      Hot Moderate
      32       34       87
>
```

3) Visualization: Create a scatter plot of Solar.R vs. Ozone. Map the "Temperature Category" to the color aesthetic.

```
> library(ggplot2)
> ggplot(clean_data, aes(x = Solar.R, y = Ozone, color = Temp_Category)) +
  geom_point()
>
```

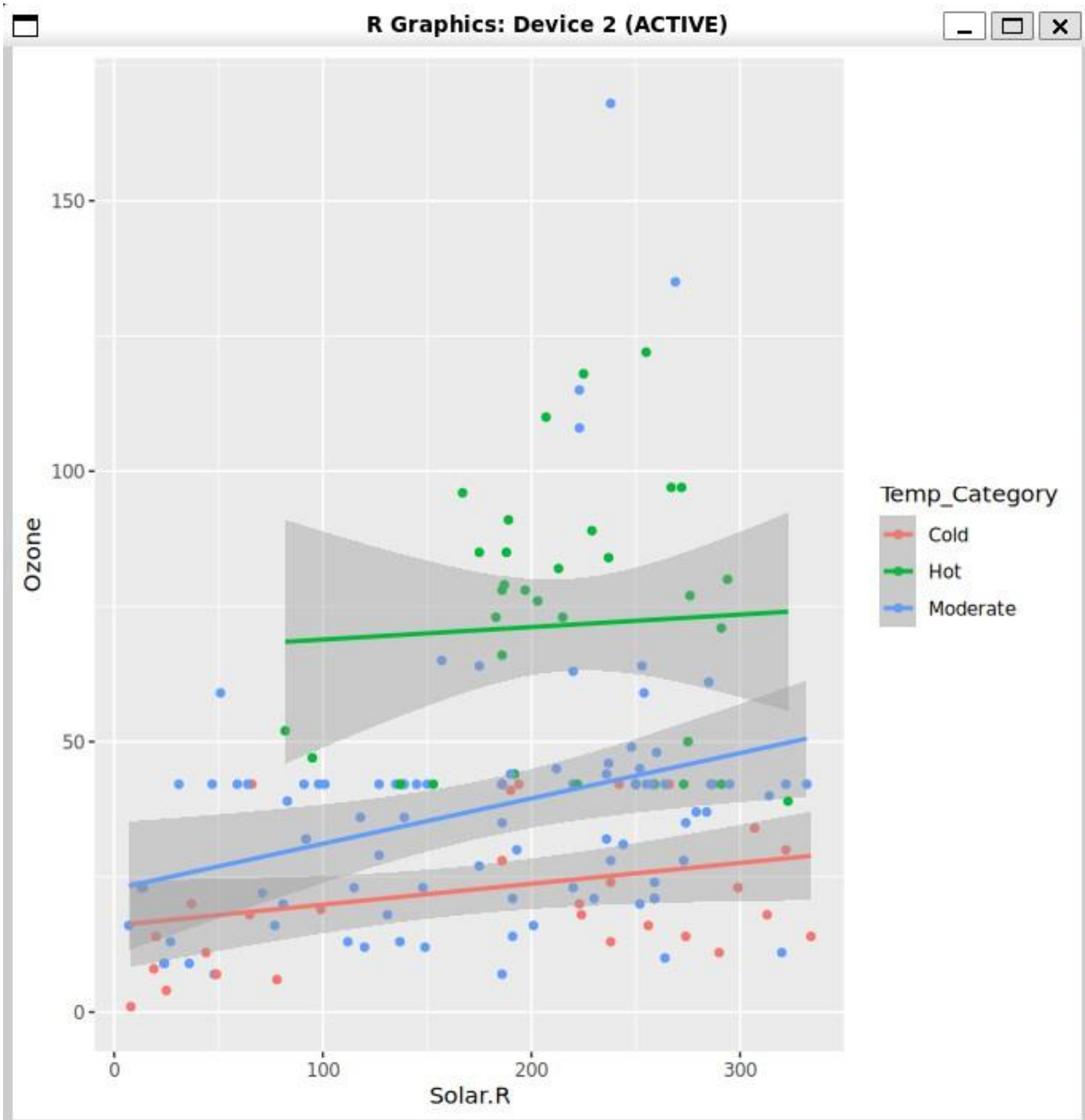
Output :




4) Requirement: Add a geom_smooth() layer to show the trend.

```
> ggplot(clean_data, aes(x = Solar.R, y = Ozone, color = Temp_Category)) +
  geom_point() +
  geom_smooth(method = "lm")
`geom_smooth()` using formula = 'y ~ x'
> |
```

Output :




**Exercise 2**: Multi-Variable Distribution
Goal: Compare distributions across categories.

- Dataset: File Name: server_logs.csv

Columns: Timestamp, Server_ID, CPU_Load (%), RAM_Usage (GB), Response_Time (ms), Error_Count.

- Tasks:

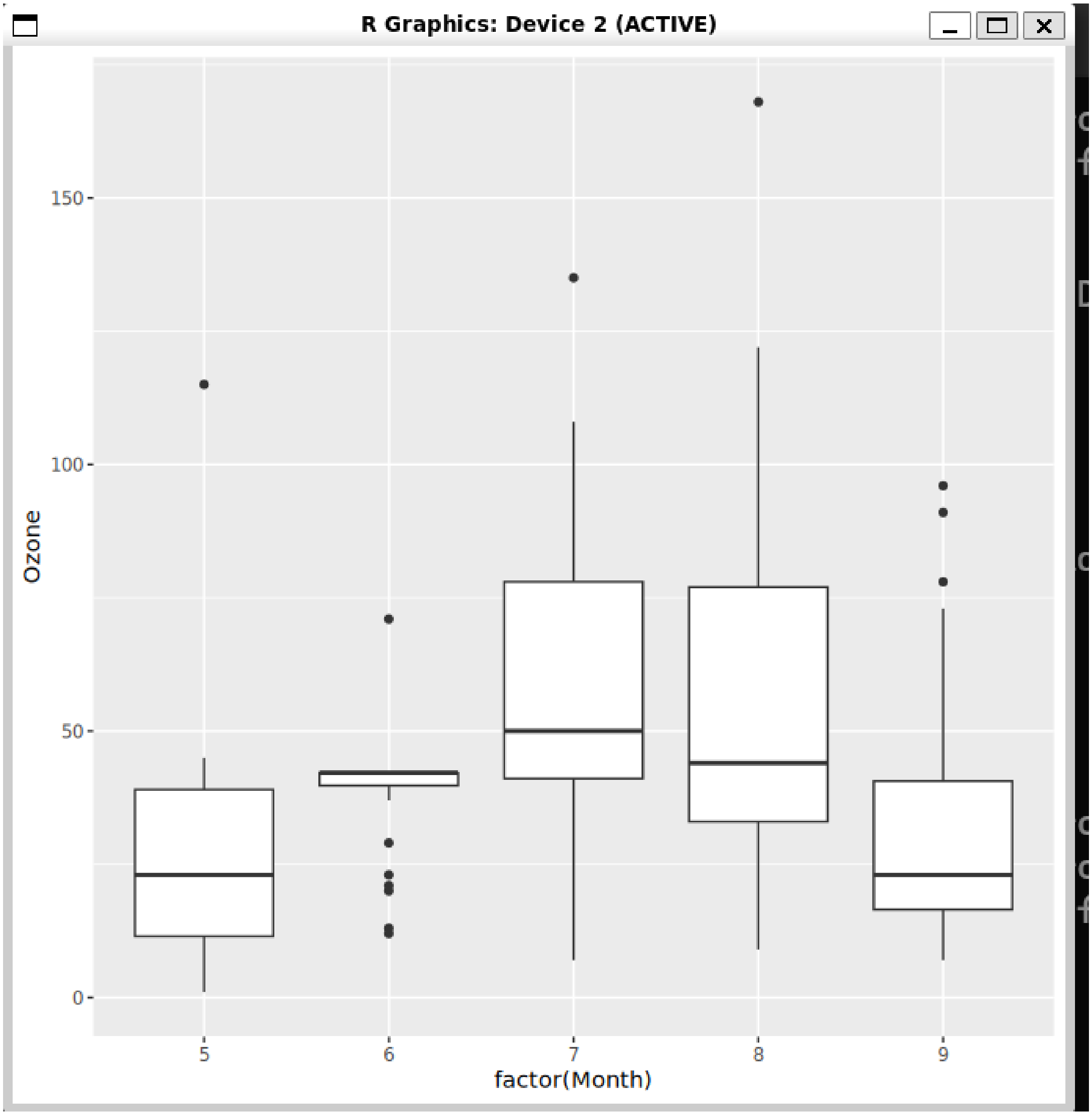
1. Create a Boxplot comparing Ozone levels across different Months.
2. Apply a custom color palette and theme (e.g., theme_minimal()).

Ans :

1) Create a Boxplot comparing Ozone levels across different Months.

```
> ggplot(clean_data, aes(x = factor(Month), y = Ozone)) +
  geom_boxplot()
> 
```

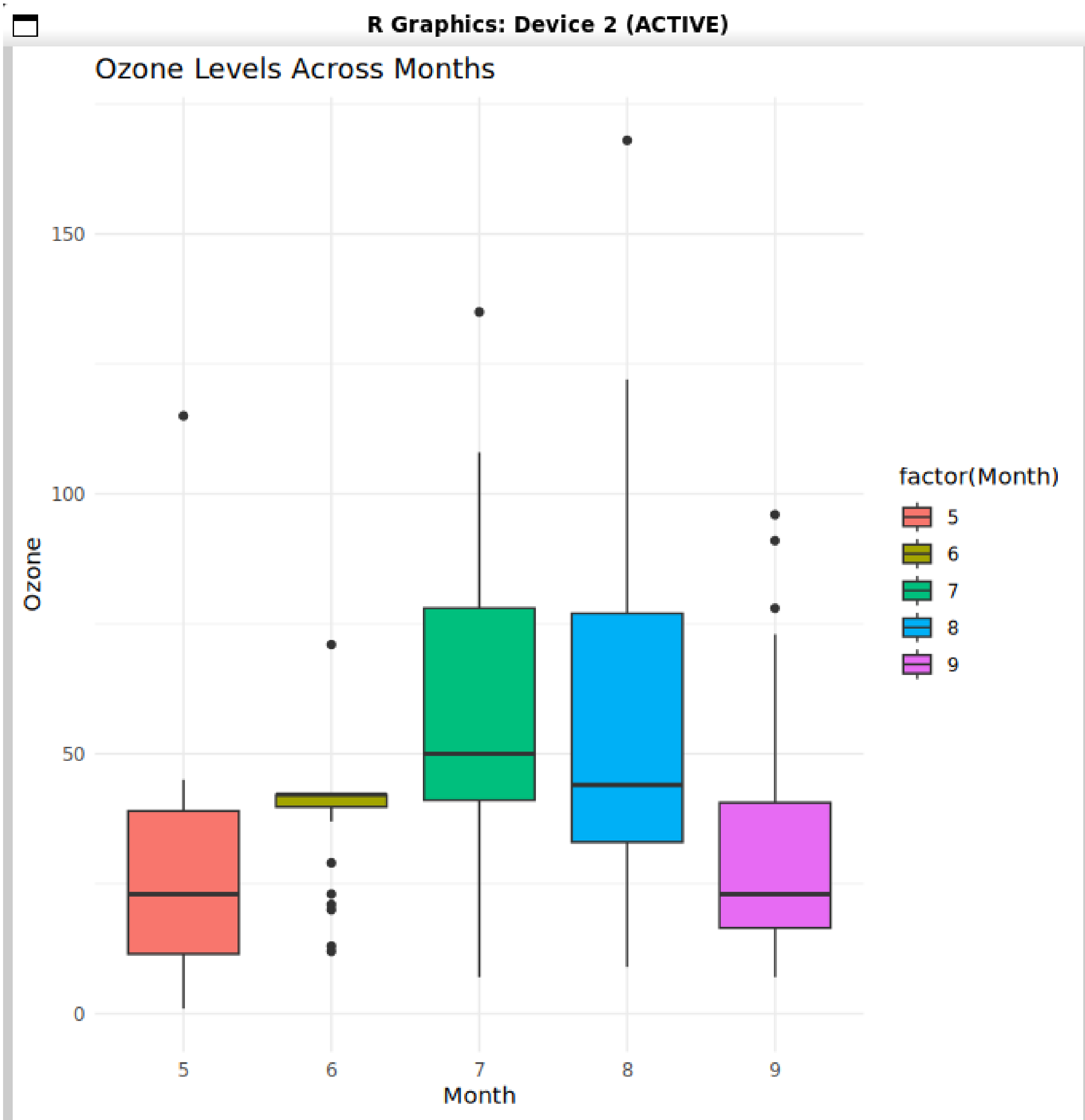
Output :




2) Apply a custom color palette and theme (e.g., theme_minimal()).

```
> ggplot(clean_data, aes(x = factor(Month), y = Ozone, fill = factor(Month))) +
  geom_boxplot() +
  theme_minimal() +
  labs(
    title = "Ozone Levels Across Months",
    x = "Month",
    y = "Ozone"
  )
> |
```

Output :


R Graphics: Device 2 (ACTIVE)
Ozone Levels Across Months
150
100
50
0
Ozone
5
6
7
8
9
Month
factor(Month)
5
6
7
8
9